MUHAMMAD RISKI ALDE

Data Analyst | Data Scientist | AI & Analytics

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EXECUTIVE SUMMARY

Applied Data Science graduate with a 3.87/4.00 GPA and hands-on experience in Data Analytics, Machine Learning, AI applications, and data-driven product development. Experienced through Data Scientist internships at PT Perusahaan Gas Negara Tbk (PGN) and PT Telekomunikasi Selular (Telkomsel), R&D analytics and business strategy work at Hayago Robotika, and applied projects across energy, telecommunications, agriculture, production, and social-impact technology. Skilled in translating business and operational problems into analytical solutions using Python, SQL, machine learning, predictive modeling, and data visualization.

EDUCATION

Politeknik Elektronika Negeri Surabaya (PENS) — Surabaya, Indonesia 2022 - 2026

- Bachelor of Applied Science in Applied Data Science (D4) | GPA: 3.87/4.00
- Relevant Coursework: Data Analysis, Machine Learning, Artificial Intelligence, Data Mining, Database Systems

WORKING EXPERIENCES

PT Hayago Robotika Indonesia (AITOMA) — Surabaya, Indonesia Jun 2026 – Aug 2026

R&D Data Analyst & Business Strategy Consultant (Freelance)

- Conducted quantitative and qualitative market validation to evaluate product-market fit for AITOMA's robotics and AI solutions, translating findings into an optimized Business Model Canvas (BMC).
- Developed financial projections, RAB budgets, and analytical pitch materials covering R&D expenditures, revenue streams, and valuation targets.
- Assessed Technology Readiness Level (TRL/TKT) and defined KPI-based output strategies to connect R&D milestones with commercial objectives.

PT Perusahaan Gas Negara Tbk (PGN) — Surabaya, Indonesia Aug 2025 – Nov 2025

Data Scientist Intern

- Built an AI-assisted document search and matching system to streamline customer verification and identify missing documents across thousands of customer records, reducing identification time from 1–2 days to 1–2 seconds.
- Integrated multi-source operational and socio-demographic data across 153 sub-districts to develop a spatial analytics system for mapping and assessing customer churn risks within the City Gas (Jargas) network.
- Developed and evaluated Gradient Boosting and SVM models with Gaussian Noise Injection to support predictive modeling under limited-data conditions.
- Deployed interactive Streamlit and Folium dashboards integrating geographic risk-zone visualization, operational business logic, and SHAP-based feature importance.

PT Telekomunikasi Selular (Telkomsel) — Surabaya, Indonesia Jan 2025 – Jun 2025

Data Scientist Intern

- Designed interactive dashboards to visualize monthly budget usage, spending patterns, and multi-level approval flows.
- Built data structures and workflows to support evidence-based tracking and reporting.
- Collaborated with cross-campus teams (ITS, PENS) to align analytical features with user needs and business goals.

SiFabel — Surabaya, Indonesia Feb 2024 - Apr 2025

CEO & CO-Founder

- Led the development of an AI-powered automatic book-reading application designed to improve literacy access for visually impaired users under PENSSKY Venture Inkubator.
- Established strategic partnerships with foundations and organizations, expanding platform reach to 100+ active visually impaired users.
- Oversaw product development, intellectual-property registration, grant activities, team coordination, and business roadmap.

SELECTED DATA & AI PROJECTS

EfiDash — Production Efficiency Smart Alert System PT Indonesia Berlian Yasawirya March 2025 – Jun 2026

Final Academic Project — Data Scientist & Developer

- Developed a production-efficiency monitoring and smart-alert platform using Dynamic Factor Analysis (DFA), feature engineering, and predictive modeling to identify operational anomalies.
- Built automated data-processing and prediction pipelines with proactive alerts and interactive dashboards to support production monitoring and decision-making.

Republik Melon (Replon Greenhouse Monitoring System) - Blitar, Indonesia Dec 2023 – Dec 2024

Dashboard Developer — Kedaireka Matching Fund

- Led the development of an IoT analytics dashboard to enhance data visibility and operational decision-making for melon farmers.
- Engineered real-time data pipelines integrating multi-node sensors to monitor greenhouse environmental conditions.
- Collaborated with cross-functional stakeholders to define KPI metrics, improving yield prediction accuracy.

Spending Scanner - Surabaya, Indonesia Feb 2024 – Jun 2024

PIC Dashboard Development

- Led the development of a financial management dashboard for the Spending Scanner application, enabling users to track expenditures effectively and gain insights into their spending patterns.
- Implemented predictive analytics features to forecast future spending trends, assisting users in making informed financial decisions.

Student Perspective Dashboard of PENS - Surabaya, Indonesia Oct 2023 – Nov 2023

PIC Dashboard Development

- Developed a comprehensive dashboard analyzing student perspectives at PENS, showcasing responses from 123 participants across various departments.
- Utilized visualizations such as pie charts and bar graphs to illustrate departmental representation and the total responses for each question, facilitating data-driven insights for academic improvements.

Data Visualization of Education Quality Indicators - Surabaya, Indonesia Sep 2023 – Oct 2023

Data Analyst

- Developed an interactive dashboard for visualizing education quality indicators across Indonesia, incorporating geographic maps and silhouette scores for effective clustering analysis.
- Implemented line charts to display annual average education trends, enabling stakeholders to monitor progress and make informed decisions regarding education policies.

Quick Count Application Implementation for Predicting - Surabaya, Indonesia Sep 2023 – Oct 2023

Data Analyst

- Led the development of a Quick Count application to predict the Malang Bupati election results using statistical modeling techniques to enhance accuracy and support decision-making.

Imbalanced Dataset Classification on Indonesia's Education Quality - Surabaya, Indonesia Sep 2023

Data Scientist

- Led a data classification project focusing on imbalanced datasets to evaluate the quality of education in Indonesia, utilizing various resampling methods for better decision-making.
- Analyzed key factors influencing school accreditation, providing insights that support more equitable education policies and contribute to SDG 4.

Waste Prediction in California - Surabaya, Indonesia Apr 2023 – Jun 2023

Data Scientist

- Developed a predictive model using the Arima Method to estimate waste production, enabling more efficient and sustainable waste management practices.
- Contributed to SDG 12 by delivering actionable insights that support responsible consumption and long-term waste reduction strategies.

ORGANIZATIONAL EXPERIENCES

HIPMI Perguruan Tinggi PENS – Surabaya, Indonesia 2024 - 2025

Head of Organizational, Membership, and Regeneration Division

• Managed organizational structure, member recruitment, training, and leadership development.

• Led initiatives to increase member quality and quantity through regeneration programs.

• Aligned all cadre programs with HIPMI's mission to foster competent young entrepreneurs.

EEPIS Data Enthusiasts 2024 - 2025

Founder

- Developed the community's long-term vision and strategy.
- Managed resources (budget, time, personnel) for sustainability.
- Built a network for collaboration and learning with academics and data communities.

Badan Perwakilan Anggota (BPA) HIMIT PENS 2024 - 2025

Legal Division

- Managed and enforced internal organizational policies and regulations.
- Provided legal advice to ensure compliance with applicable laws.
- Handled legal issues arising from organizational activities.

PMII Wira Terapan 1011 2024 - 2025

Internal Development Division

- Developed and evaluated programs to improve cadre quality and organizational performance, including training and curriculum development.
- Facilitated organizational change and promoted a positive, innovative culture to support organizational goals.

Forum Bidikmisi Kabupaten Ponorogo 2023 - 2025

Networking Division

- Built networks, coordinated events (GTS, Podcast, Gathering, and 4BMP Berprestasi) and ensured program success.
- Developed strategies for long-term growth and improved program effectiveness.

Dewan Kerja Cabang (DKC) Ponorogo 2022 - 2024

Head of Scouting Research

- Led scouting research, program development, and innovation.
- Facilitated learning through discussions, workshops, and presentations.
- Developed curricula and training materials based on current trends and needs.

SELECTED AWARDS & ACHIEVEMENTS

- **1st Place** - National Startup Competition (NSC 2024), Universitas Negeri Surabaya **2024**
- **1st Place** - Business Collaboration, Pameran Inovasi & Teknologi Vokasi (PENSASI 2024), PENSSKY Venture **2024**
- **Finalist & Beneficiary** - Program Pembinaan Mahasiswa Wirausaha (P2MW), Kemendikbud RI **2024**
- **Finalist** - GEMASTIK XVII, ICT Business Development Division, Puspresnas **2024**
- **Recipient** - Program Kreativitas Mahasiswa Artikel Ilmiah (PKM-AI), Kemendikbud RI **2024**
- **3rd Place** - National Scientific Writing Competition (LKTI), Universitas Riau **2024**

PATENTS & INTELLECTUAL PROPERTY (HKI)

- **Copyright / HKI** - Buku Manual Aplikasi Petani Greenhouse Melon Di Kecamatan Wates Kabupaten Blitar **2024**
- **Copyright / HKI** - Buku Manual Aplikasi RepLon Dashboard Untuk Monitoring Dan Prediksi Kondisi Greenhouse Melon Multi-Node Berbasis Data Mining **2024**
- **Copyright / HKI** - Aplikasi Chatbot Informasi Pelayanan Kelurahan Keputih "BambuBot" **2024**
- **Copyright / HKI** - Aplikasi Pembaca Buku Otomatis Berbasis Artificial Intelligence Untuk Meningkatkan **2024**

Kemandirian Dan Akses Literasi Tunanetra

RESEARCH & PUBLICATIONS

- **El-Mujtama Journal (Main Author / Developer)** **2024**
  Implementasi Aplikasi Chatbot Informasi Pelayanan Kelurahan Keputih, Surabaya (BambuBot)

- **Kedaireka Matching Fund Project** **2024**
  Pengembangan Dashboard Kondisi Greenhouse Multi-Node Berbasis Data Mining Sebagai Upaya Pemberdayaan Petani Melon Kecamatan Wates Kabupaten Blitar

TECHNICAL SKILLS

- **Programming & Data**: Python, SQL, R, Excel, Data Mining, Machine Learning, Dynamic Factor Analysis (DFA), Spatial Data Analysis
- **Analytics & Visualization**: Tableau, Power BI, Looker Studio, Streamlit, Folium, Grafana, Minitab
- **AI & Modeling**: Artificial Intelligence, Predictive Modeling, SHAP, Classification, Forecasting
- **R&D & Business Analytics**: Market Validation, TKT/TRL Evaluation, Financial Modeling, RAB Budgeting, Business Model Canvas
- **Languages**: Bahasa Indonesia (Native), English